



Java Tutorial 1.4

Q1	Write a Java program to store 10 integers in an array and print only the prime numbers from it.
Code	<pre>class PrimeInArray { public static void main(String[] args) { int[] arr = {12, 5, 7, 19, 24, 31, 40, 43, 50, 53}; for (int i = 0; i < arr.length; i++) { int c = 0; for (int j = 1; j <= arr[i]; j++) { if (arr[i] % j == 0) { c++; } } if (c == 2) { System.out.println(arr[i]); } } } }</pre>
Output	<pre>D:\WorkSpaces\Java\Assignments\Tut_1_4>java PrimeInArray 5 7 19 31 43 53</pre>
Q2	Write a program to find the second largest and second smallest element in an array without using sorting.
Code	<pre>class SecondMinMax { public static void main(String[] args) {</pre>



	<pre>int[] arr = {15, 2, 8, 20, 5, 12}; int max = Integer.MIN_VALUE; int secondMax = Integer.MIN_VALUE; int min = Integer.MAX_VALUE; int secondMin = Integer.MAX_VALUE; for (int i = 0; i < arr.length; i++) { if (arr[i] > max) { secondMax = max; max = arr[i]; } else if (arr[i] > secondMax && arr[i] != max) { secondMax = arr[i]; } if (arr[i] < min) { secondMin = min; min = arr[i]; } else if (arr[i] < secondMin && arr[i] != min) { secondMin = arr[i]; } } System.out.println("Second Largest: " + secondMax); System.out.println("Second Smallest: " + secondMin); }</pre>
Output	<pre>D:\WorkSpaces\Java\Assignments\Tut_1_4>java SecondMinMax Second Largest: 15 Second Smallest: 5</pre>
Q3	Write a program to check whether an array is a palindrome array or not.
Code	<pre>class PalindromeArray { public static void main(String[] args)</pre>



	<pre>{ int[] arr = {1, 2, 3, 2, 1}; boolean isPalindrome = true; int n = arr.length; for (int i = 0; i < n / 2; i++) { if (arr[i] != arr[n - 1 - i]) { isPalindrome = false; break; } } if (isPalindrome) { System.out.println("Array is a palindrome"); } else { System.out.println("Array is not a palindrome"); } }</pre>
Output	<pre>D:\WorkSpaces\Java\Assignments\Tut_1_4>java PalindromeArray Array is a palindrome</pre>
Q4	Write a Java program to accept marks of N students (user input size) and display the highest, lowest, and average marks.
Code	<pre>import java.util.Scanner; class StudentMarks { public static void main(String[] args) { Scanner sc = new Scanner(System.in); System.out.print("Enter number of students: "); int n = sc.nextInt(); int[] marks = new int[n]; int sum = 0;</pre>



	<pre>int highest = Integer.MIN_VALUE; int lowest = Integer.MAX_VALUE; for (int i = 0; i < n; i++) { System.out.print("Enter marks for student " + (i + 1) + ": "); marks[i] = sc.nextInt(); sum += marks[i]; if (marks[i] > highest) { highest = marks[i]; } if (marks[i] < lowest) { lowest = marks[i]; } } double average = (double) sum / n; System.out.println("Highest Marks: " + highest); System.out.println("Lowest Marks: " + lowest); System.out.println("Average Marks: " + average); }</pre>
Output	<pre>D:\WorkSpaces\Java\Assignments\Tut_1_4>java StudentMarks Enter number of students: 3 Enter marks for student 1: 12 Enter marks for student 2: 34 Enter marks for student 3: 54 Highest Marks: 54 Lowest Marks: 12 Average Marks: 33.333333333333336</pre>
Q1	Write a program to print the reverse multiplication table of a given number (from 10 to 1).
Code	<pre>import java.util.Scanner; class ReverseTable</pre>



	<pre>{ public static void main(String[] args) { Scanner sc = new Scanner(System.in); System.out.print("Enter a number: "); int num = sc.nextInt(); for (int i = 10; i >= 1; i--) { System.out.println(num + " x " + i + " = " + (num * i)); } } }</pre>
Output	<pre>D:\WorkSpaces\Java\Assignments\Tut_1_4>java ReverseTable Enter a number: 3 3 x 10 = 30 3 x 9 = 27 3 x 8 = 24 3 x 7 = 21 3 x 6 = 18 3 x 5 = 15 3 x 4 = 12 3 x 3 = 9 3 x 2 = 6 3 x 1 = 3</pre>
Q2	Write a program to calculate the factorial of a number using recursion and loop (compare both).
Code	<pre>import java.util.Scanner; class FactorialCompare { static int factRec(int n) { if (n == 0 n == 1) { return 1; } return n * factRec(n - 1); } }</pre>



	<pre>public static void main(String[] args) { Scanner sc = new Scanner(System.in); System.out.print("Enter a number: "); int n = sc.nextInt(); int factLoop = 1; for (int i = 1; i <= n; i++) { factLoop *= i; } int factRecursive = factRec(n); System.out.println("Factorial using Loop: " + factLoop); System.out.println("Factorial using Recursion: " + factRecursive); } }</pre>
Output	<pre>D:\WorkSpaces\Java\Assignments\Tut_1_4>java FactorialCompare Enter a number: 3 Factorial using Loop: 6 Factorial using Recursion: 6</pre>
Q3	Write a program to check whether a given number is palindrome using a loop.
Code	<pre>import java.util.Scanner; class PalindromeNumber { public static void main(String[] args) { Scanner sc = new Scanner(System.in); System.out.print("Enter a number: "); int num = sc.nextInt(); int original = num; int reverse = 0; while (num != 0) { int digit = num % 10; reverse = reverse * 10 + digit;</pre>



	<pre> num /= 10; } if (original == reverse) { System.out.println("Palindrome number"); } else { System.out.println("Not a palindrome number"); } } }</pre>
Output	<pre>D:\WorkSpaces\Java\Assignments\Tut_1_4>java PalindromeNumber Enter a number: 77 Palindrome number</pre>
Q4	Write a program to print a below pattern using nested loops.
Code	<pre>class PatternDisplay { public static void main(String[] args) { for (int i = 1; i <= 3; i++) { for (int j = 1; j <= 4 - i; j++) { System.out.print(" "); } for (int k = 1; k <= (2 * i - 1); k++) { System.out.print("*"); } System.out.println(); } for (int i = 1; i <= 7; i++) { System.out.print(i); } System.out.println(); for (int i = 3; i >= 1; i--)</pre>



	<pre>{ for (int j = 1; j <= 4 - i; j++) { System.out.print(" "); } for (int k = 1; k <= (2 * i - 1); k++) { System.out.print("*"); } System.out.println(); } }</pre>
Output	<pre>D:\WorkSpaces\Java\Assignments\Tut_1_4>java PatternDisplay * *** ***** 1234567 ***** *** *</pre>
Q5	Write a program to implement matrix addition and multiplication using 2D arrays.
Code	<pre>class MatrixOperations { public static void main(String[] args) { int[][] a = {{1, 2}, {3, 4}}; int[][] b = {{5, 6}, {7, 8}}; int[][] sum = new int[2][2]; int[][] prod = new int[2][2]; for (int i = 0; i < 2; i++) { for (int j = 0; j < 2; j++) { sum[i][j] = a[i][j] + b[i][j]; } } for (int i = 0; i < 2; i++)</pre>



	<pre>{ for (int j = 0; j < 2; j++) { prod[i][j] = 0; for (int k = 0; k < 2; k++) { prod[i][j] += a[i][k] * b[k][j]; } } } System.out.println("Addition:"); for (int i = 0; i < 2; i++) { for (int j = 0; j < 2; j++) { System.out.print(sum[i][j] + " "); } System.out.println(); } System.out.println("Multiplication:"); for (int i = 0; i < 2; i++) { for (int j = 0; j < 2; j++) { System.out.print(prod[i][j] + " "); } System.out.println(); } }</pre>
Output	<pre>D:\WorkSpaces\Java\Assignments\Tut_1_4>java MatrixOperations Addition: 6 8 10 12 Multiplication: 19 22 43 50</pre>



Q1	Write a Java program to create a jagged array of 3 rows (2, 3, 4 elements). Fill values from the user and print the jagged array.
Code	<pre>import java.util.Scanner; class JaggedArrayBasic { public static void main(String[] args) { Scanner sc = new Scanner(System.in); int[][] arr = new int[3][]; arr[0] = new int[2]; arr[1] = new int[3]; arr[2] = new int[4]; for (int i = 0; i < arr.length; i++) { for (int j = 0; j < arr[i].length; j++) { System.out.print("Enter value for row " + i + " col " + j + ": "); arr[i][j] = sc.nextInt(); } } System.out.println("Jagged Array Elements:"); for (int i = 0; i < arr.length; i++) { for (int j = 0; j < arr[i].length; j++) { System.out.print(arr[i][j] + " "); } System.out.println(); } } }</pre>



Output	<pre>D:\WorkSpaces\Java\Assignments\Tut_1_4>javac JaggedArrayBasic.java D:\WorkSpaces\Java\Assignments\Tut_1_4>java JaggedArrayBasic Enter value for row 0 col 0: 77 Enter value for row 0 col 1: 88 Enter value for row 1 col 0: 99 Enter value for row 1 col 1: 11 Enter value for row 1 col 2: 22 Enter value for row 2 col 0: 33 Enter value for row 2 col 1: 44 Enter value for row 2 col 2: 55 Enter value for row 2 col 3: 66 Jagged Array Elements: 77 88 99 11 22 33 44 55 66</pre>
Q2	Suppose 3 students appeared in different numbers of subjects. Use a jagged array to store marks and print each student's marks.
Code	<pre>import java.util.Scanner; class StudentMarksJagged { public static void main(String[] args) { Scanner sc = new Scanner(System.in); int[][] marks = new int[3][]; marks[0] = new int[2]; marks[1] = new int[4]; marks[2] = new int[3]; String[] students = {"Hardik", "Krunal", "Kishan"}; for (int i = 0; i < marks.length; i++) { System.out.println("Enter marks for " + students[i] + ":"); for (int j = 0; j < marks[i].length; j++) { System.out.print("Subject " + (j + 1) + ": "); marks[i][j] = sc.nextInt(); } } System.out.println(); } }</pre>



	<pre>for (int i = 0; i < marks.length; i++) { System.out.print(students[i] + "'s marks: "); for (int j = 0; j < marks[i].length; j++) { System.out.print(marks[i][j] + " "); } System.out.println(); } }</pre>
Output	<pre>D:\WorkSpaces\Java\Assignments\Tut_1_4>javac StudentMarksJagged.java D:\WorkSpaces\Java\Assignments\Tut_1_4>java StudentMarksJagged Enter marks for Hardik: Subject 1: 23 Subject 2: 45 Enter marks for Krunal: Subject 1: 56 Subject 2: 34 Subject 3: 23 Subject 4: 54 Enter marks for Kishan: Subject 1: 56 Subject 2: 67 Subject 3: 76 Hardik's marks: 23 45 Krunal's marks: 56 34 23 54 Kishan's marks: 56 67 76</pre>
Q3	Write a program to store elements in a jagged array and find the maximum element from the entire jagged array.
Code	<pre>class JaggedArrayMax { public static void main(String[] args) { int[][] arr = { {12, 45}, {9, 88, 34}, {56, 99, 21, 10} }; int max = arr[0][0];</pre>



	<pre>for (int i = 0; i < arr.length; i++) { for (int j = 0; j < arr[i].length; j++) { if (arr[i][j] > max) { max = arr[i][j]; } } } System.out.println("Maximum element in the jagged array: " + max); }</pre>
Output	<pre>D:\WorkSpaces\Java\Assignments\Tut_1_4>javac JaggedArrayMax.java D:\WorkSpaces\Java\Assignments\Tut_1_4>java JaggedArrayMax Maximum element in the jagged array: 99</pre>
Q4	Write a program to print a below pattern using jagged array.
Code	<pre>class JaggedArrayPattern { public static void main(String[] args) { int[][] arr = new int[4][]; for (int i = 0; i < 4; i++) { arr[i] = new int[i + 1]; for (int j = 0; j <= i; j++) { arr[i][j] = j + 1; } } for (int i = 0; i < arr.length; i++) { for (int j = 0; j < arr[i].length; j++) { System.out.print(arr[i][j] + " "); } } } }</pre>



	<pre> System.out.println(); } } }</pre>
Output	<pre>D:\WorkSpaces\Java\Assignments\Tut_1_4>java JaggedArrayPattern 1 1 2 1 2 3 1 2 3 4</pre>